

The Likelihood Ratio Test

To this point we have considered cases where the choice of the test statistic was either (1) based on “intuition,” i.e. an understanding of what statistic would be relevant in assessing a particular hypothesis; or (2) provided to us, e.g., in the Phillips-Ouliaris Test.

There is a general framework, based on maximum likelihood, for constructing test statistics that are optimal in the sense that they control the probability of Type II error as much as possible for fixed choice of α . Crucially, the test statistic that results has a simple (approximate) distribution under the null hypothesis; we have already seen how important this is to performing a test.

The test is referred to as the **likelihood ratio test** because it constructs a test statistic for testing $H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \notin \Theta_0$ via the ratio

$$T = -2 \log \left(\frac{\max_{\theta \in \Theta_0} L(\theta)}{\max_{\theta \notin \Theta_0} L(\theta)} \right)$$

Exercise: Explain why this is a reasonable choice of test statistic for performing this test. For what values of T should this test reject?

If $\frac{\max_{\theta \in \Theta_0} L(\theta)}{\max_{\theta \notin \Theta_0} L(\theta)}$ is small we have evidence against $H_0 : \theta \in \Theta_0$ and in favor of $H_1 : \theta \notin \Theta_0$.
 $T = -2 \log$ get larger as the ratio gets smaller. Large values of T should lead to a rejection.

Important Result: Under “regularity conditions” and the assumption that $P(T = 0) = 0$, then T has (asymptotically) the chi-squared distribution with degrees of freedom equal to

Number of free parameters under H_1 – Number of free parameters under H_0

e.g. $H_0: \mu_1 = \mu_2$ vs. $H_1: \mu_1 \neq \mu_2$, so $T \sim \chi^2_{2-1}$ under H_0

Hence, for fixed choice of the level α , one would reject H_0 if T is larger than $\text{chi2.ppf}(1-\alpha, \text{df})$. Alternatively, the p-value can be calculated as $1 - \text{chi2.cdf}(T, \text{df})$.



Exercise: Is the assumption that $P(T = 0) = 0$ restrictive in practice?

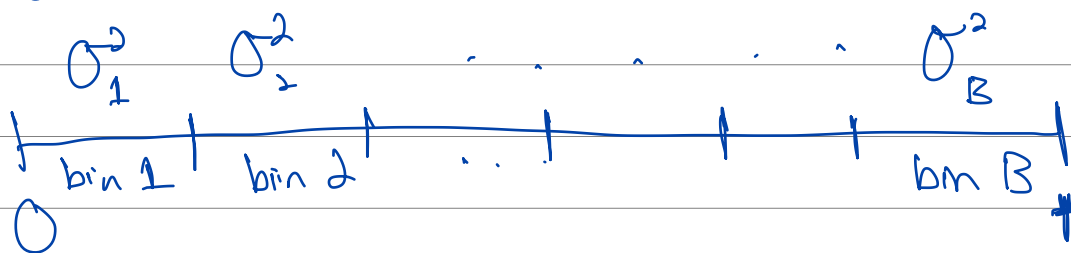
Exercise: Develop a test for constant volatility.

Assume observe $X_1, X_2, \dots, X_n$, the log returns over each of n equally-spaced time intervals

Could assume $X_1, X_2, \dots, X_n$ independent Normal, $E(X_i) = 0$

Let $V(X_i) = \sigma_i^2$. Under H_0 , $\sigma_i^2 = \sigma^2$ for all i .
 Could define H_1 such that
under H_1 , σ_i^2 is not constant.

Or: Could divide the time interval into B bins. Model assumes constant vol within a bin.



$H_1: \sigma_i^2 = \sigma_b^2$ when obs. i belongs to the b^{th} bin

$H_0: \sigma_1^2 = \sigma_2^2 = \dots = \sigma_B^2$

Use LRT.

Here: $\underline{\theta} = (\sigma_1^2, \dots, \sigma_B^2)$

$$H_0: \sigma_1^2 = \sigma_2^2 = \dots = \sigma_B^2$$

$$H_1: \sigma_i^2 \neq \sigma_j^2 \text{ for some } i \neq j$$

Under H_0 , likelihood maximized when

$$\hat{\sigma}_1^2 = \hat{\sigma}_2^2 = \dots = \hat{\sigma}_B^2 = \sum X_i^2 / n$$

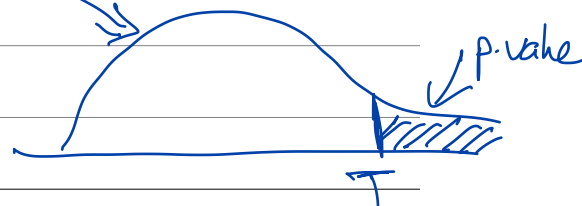
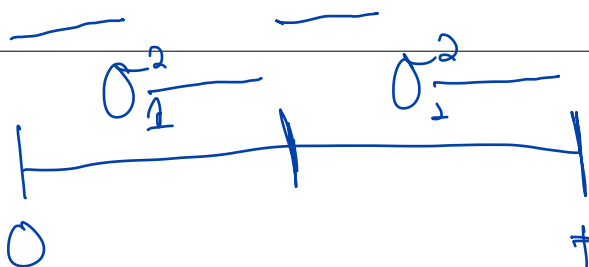
Under H_1 , likelihood maximized when

$$\hat{\sigma}_b^2 = \sum_{\substack{j: X_j \\ \text{in } b^{\text{th bin}}}} X_j^2 / n_b \leftarrow \# \text{ obs. in bin } b$$

Recall: $L(\underline{\theta}) = \prod_{i=1}^n f_{X_i}(x_i) \leftarrow N(0, \sigma_i^2)$

Have ingredients to find T

Under H_0 , $T \sim \chi_{B-1}^2$



$$\begin{aligned} \text{power} &= 1 - P(\text{Type II error}) \\ &= P(\text{reject } H_0 | H_1 \text{ true}) \end{aligned}$$

Part 20: Bayesian Estimation

Summary

To this point, our discussion has focused on the **frequentist** or **classical** approach to estimation and hypothesis testing. In that framework, consideration is given to how procedures perform in repeated use. Indeed, the performance of estimators can be assessed without using any actual data. Some find this troubling.

An alternative philosophy is the **Bayesian** approach to estimation. The central characteristic is that uncertainty in parameters is quantified using probability distributions.

As the name implies, Bayes' Rule plays a central role.

First, recall Bayes' Rule: If $B_1, B_2, \dots, B_m$ are a partition of Ω , then

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_j P(A|B_j)P(B_j)}$$

Also recall the continuous version:

$$f_{Y|X}(y|x) = \frac{f_{X|Y}(x|y)f_Y(y)}{\int f_{X|Y}(x|u)f_Y(u) du}$$

Here we will use this result with the parameter θ "playing the role" of Y :

$$\pi(\theta|\mathbf{x}) = \frac{f(\mathbf{x}|\theta)\pi(\theta)}{\int f(\mathbf{x}|t)\pi(t) dt}$$

The vector $\mathbf{x}$ is observed data.

The density $\pi(\theta)$ is the **prior distribution** for θ .

The conditional density $\pi(\theta|\mathbf{x})$ is the **posterior distribution** for θ .

- 1 In the Bayesian framework, the prior reflects the someone's **opinion** re-
- 2 garding the value of θ , **prior to** the observation of the data.
- 3 The posterior is this opinion "updated" to take the data into consideration.
- 4 *How is the prior chosen?* This can be tricky, and subjective.
- 5 The use of the Bayesian approach was long limited by computational diffi-
- 6 culties associated with calculating the required integral. With the advent of
- 7 **Markov Chain Monte Carlo (MCMC)** algorithms, these issues have largely
- 8 gone away, and the use of Bayesian methods has grown greatly.
- 9 In some relatively simple cases, analytical solutions are possible.



$$EMX = \text{MCMC}$$

$$E = MC^2$$

- 1 **Exercise:** Suppose that the observable data are modelled by i.i.d.
- 2 Bernoulli(θ) random variables $X_1, X_2, \dots, X_n$. You quantify your prior
- 3 beliefs regarding θ by using the Beta(α, β) distribution, for some chosen
- 4 values of α and β . Derive the posterior for θ in this case.

First, recall the Beta dist:

$$\pi(\theta) = \left(\frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \right) \theta^{\alpha-1} (1-\theta)^{\beta-1}, \quad 0 < \theta < 1$$

The dist. for the data

$$f(\underline{x} | \theta) = \prod_{i=1}^n f(x_i | \theta) = \prod_{i=1}^n \theta^{x_i} (1-\theta)^{1-x_i}$$

$$= \theta^{\sum x_i} (1-\theta)^{n - \sum x_i}$$

Now use Bayes' Rule

$$\pi(\theta | \underline{x}) = \frac{f(\underline{x} | \theta) \pi(\theta)}{\int f(\underline{x} | t) \pi(t) dt} = k_1 f(\underline{x} | \theta) \pi(\theta)$$

where k_1 does not depend on θ

$$= k_1 \left[\theta^{\sum x_i} (1-\theta)^{n-\sum x_i} \right] \left[\frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1} \right]$$

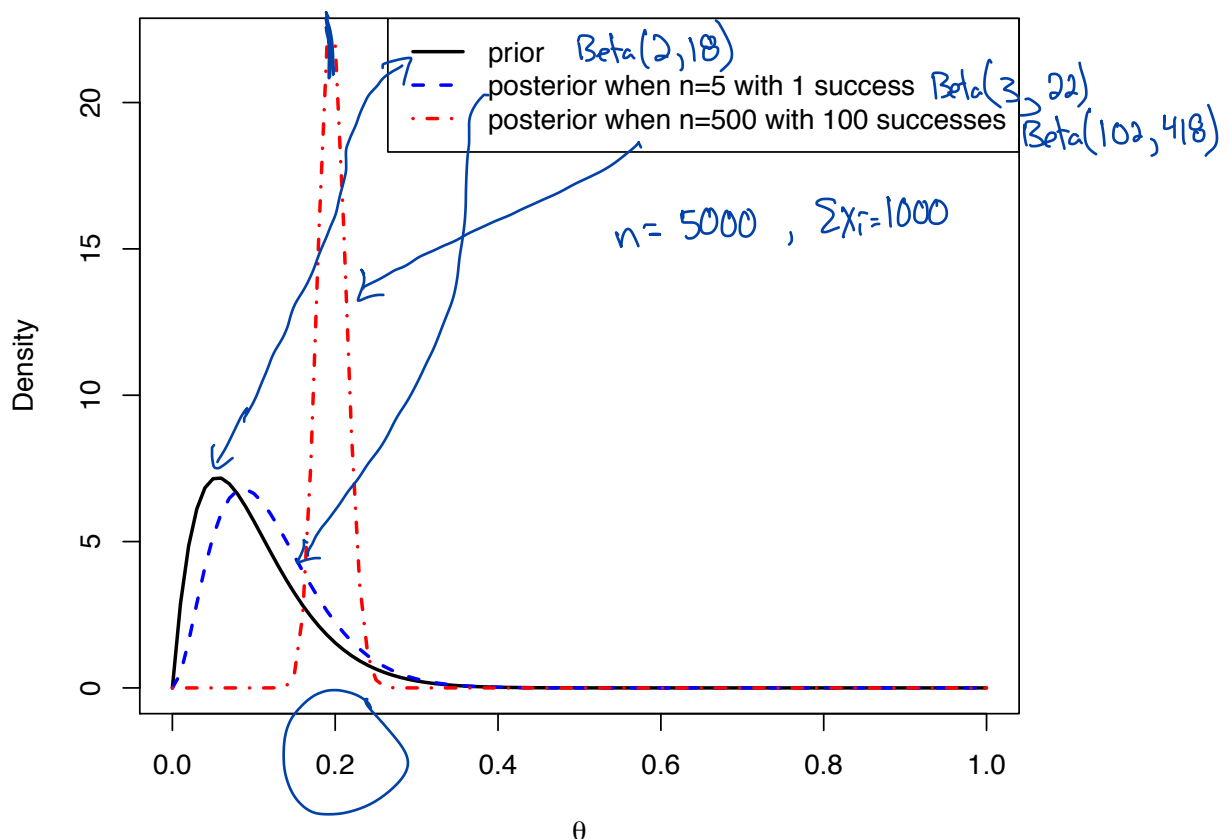
$$= k_2 \theta^{\sum x_i + (\alpha-1)} (1-\theta)^{n-\sum x_i + \beta-1}, \quad \underline{0 < \theta < 1}$$

where k_2 does not depend on θ

Can conclude that the posterior is

$$\text{Beta}(\underline{\alpha + \sum x_i}, \underline{\beta + n - \sum x_i})$$

- 1 In this example, we see that the prior and posterior are from the same
- 2 family of distributions (i.e., they are both Beta). We say that "The Beta
- 3 distribution is conjugate to sampling from the Bernoulli."
- 4 Let's consider a couple specific examples. Suppose I chose $\alpha = 2$ and
- 5 $\beta = 18$ to capture my prior opinion regarding θ . I observe $n = 5$ trials, and
- 6 $\sum_i x_i = 1$. The result above states that the posterior is Beta($2+1$, $18+(5-1)$).
 $\alpha + \sum x_i$ $\beta + n - \sum x_i$
- 7 If instead, $n = 500$ and $\sum_i = 100$, the posterior is Beta(102, 418).
- 8 The figures on the following slide compare these cases.



- 1 **Exercise:** Comment on the shapes of the posteriors in the previous exam-
 2 ple, as compared to the prior. What is the role of the sample size?

3 With small sample, the posterior is similar to
 4 the prior. As sample size grows, the data
 5 "swamps" the prior

- 1 The Bayesian often wants to convert the posterior into an estimate for θ .
 2 One way to do this is to find the maximum of the posterior, called the
 3 **maximum a posteriori (MAP) estimator**.

- 4 In our example this comes out to be

$$\frac{\alpha + \sum x_i - 1}{\alpha + \beta + n - 4}$$

- 6 Another approach is to find the posterior mean, called the **Bayes estimator**.

- 7 In our example the Bayes estimator is

$$E(\theta|\mathbf{x}) = \frac{\alpha + \sum x_i}{\alpha + \beta + n} = \left(\frac{\sum x_i}{n} \right) \left(\frac{n}{\alpha + \beta + n} \right) + \left(\frac{\alpha}{\alpha + \beta} \right) \left(\frac{\alpha + \beta}{\alpha + \beta + n} \right)$$

sample prop. prior mean

- 9 In other words, it is a weighted average of the sample proportion and the
 10 prior mean. The weight placed on the sample proportion grows with n .

- 1 **Exercise:** Return to our Bernoulli example. Choose $\alpha = \beta = 1$ for the prior.
2 What is the appeal of this choice? What is the Bayes estimator?

3 If $\alpha = \beta = 1$, uniform $(0,1)$ prior, "flat prior"

$$\begin{aligned} 4 \quad E(\theta | \underline{x}) &= \left(\frac{\sum x_i}{n} \right) \left(\frac{n}{n+2} \right) + \left(\frac{1}{2} \right) \left(\frac{2}{n+2} \right) \\ 5 \quad & \\ 6 \quad & \\ 7 \quad & \neq \sum x_i / n \end{aligned}$$

9 "There is no uninformative prior"

- 1 **Exercise:** The observable data are modelled using the Poisson(θ) distribu-
2 tion. Use the Gamma(α, β) distribution as the prior on θ . Find the posterior
3 distribution and the Bayes estimator.

1 Higher Dimensional Parameters

2 The Bayesian approach can be extended to multi-dimensional parameters.
3 The difficulties are more computational than conceptual. It is possible,
4 for instance, to place prior distributions on covariance matrices with tools
5 tailored to this purpose. Choosing priors can be even trickier in higher
6 dimensions, however.

7 For example, the paper “Bayesian Portfolio Analysis” by Avramov and
8 Zhou (*Ann. Rev. of Financial Economics*, Vol. 2) provides a nice overview of
9 Bayesian tools for selecting portfolios. Consider Equation (8) for the un-
10 derlying optimization problem. Note how uncertainty in the mean vector
11 μ and covariance matrix V are taken into consideration: Fixed estimates
12 are not plugged in, instead one integrates over the posterior for these quan-
13 tities.